

Karyotype of the Indian Spiny Mouse Resulted from Tandem Fusion of Some of the House Mouse Chromosomes¹

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It has been shown by Gropp *et al.* (1970) that the karyotype of the tobacco mouse (*Mus poschiavinus*) ($2n=26$), which consists of 7 large metacentric autosome pairs and the remaining acrocentric ones, has resulted from Robertsonian fusion of some acrocentric pairs occurring in the house mouse (*Mus musculus*). The chromosome number of the Indian spiny mouse, *Mus platythrix*, was also $2n=26$, and all chromosome pairs are acrocentrics, some being longer than those found in the house mouse (Satya Prakash and Aswathanarayana 1972, Tsuchiya and Yosida 1972, Dhanda *et al.* 1973 and Yosida 1978). On the basis of the measurement of chromosome length as well as of the G-band analysis between the spiny mouse and the house mouse, it has been assumed that several chromosome pairs of the former species were resulted from tandem fusion in each two acrocentric pairs of the latter species. Taking the karyotype evolution by tandem chromosome fusion into account, the karyotype constitution of the Indian spiny mouse is analysed in this study in comparison with the chromosomes of the house mouse, use being made of G-, C- and Ag-NOR procedures.

Material and method

The Indian spiny mice, *Mus platythrix*, which were collected in Mysore, India in 1972, and then have been bred in our institute for the last six years, were used for the present study. As the material for the house mouse, the B10A strain of the laboratory stock was used. Chromosomes were observed based on bone marrow as well as on lung cultures. The trypsin G-banding and $\text{Ba}(\text{OH})_2$ C-banding method (Yosida and Sagai 1975), and the Ag-AS nucleolar organizer technique (Goodpasture and Bloom 1975) were applied to the chromosome analyses.

Results

I. Karyotype of the Indian spiny mouse

1) Karyotype by the conventional method: The chromosome number of the Indian spiny mouse was $2n=26$, all of which were acrocentrics of varying in size (Fig. 1). The total length of all chromosome pairs (haploid set with X and Y) was 72.0μ , the longest pair (no. 1) being 7.5μ and the smallest one (no. 12) 2.5μ

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(Table 1). The length of the pair no. 2 was slightly shorter than the pair no. 1. A short knob was sometimes demonstrated in pairs 5 and 8 (Fig. 2A). The length of the X chromosome was 4.8μ , showing the length between pairs 8 and 9. The Y was 3.1μ in length lying between pairs 11 and 12.

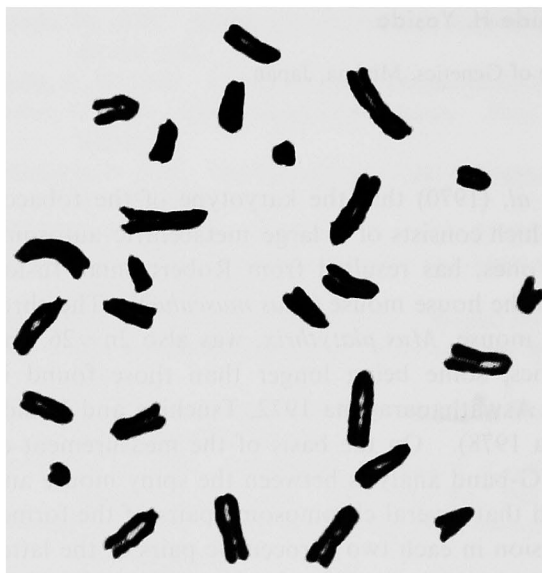


Fig. 1. Metaphase chromosomes of the Indian spiny mouse (*Mus platythrix*) ($2n=26$).

the latter. Pair no. 3 showed one dark band at a position about one third from the proximal end and one light band at the distal end. No such bands were present in pair no. 4. Pair no. 5 had a remarkable dark band at the centromeric portion together with three bands on the longer arm. The bands in pairs no. 6 and 7 were similar in appearance. But the former showed two closely located dark bands, while one dark band was present in the latter. Pair no. 8 had two dark bands separately located at the proximal and distal ends. Pair no. 9 was noted by 4 dark bands in the middle part. Pairs 10 and 11 were identified by having two and one dark bands, respectively. In the smallest pair no. 12 showed no dark band. The X chromosome was remarkable by light bands, together with a least 3 dark bands in its middle part. Proximal and distal ends of the X were also stained slightly. The Y chromosome was generally stained along its length, but two darkly stained segments were seen at its proximal part.

3) C-banding patterns: In the house mouse C-bands were clearly observed in the centromeric region of almost all chromosome pairs. In the spiny mouse, however, the bands were generally small and obscure, although tiny centromeric C-bands were recognizable in all chromosome pairs. An interstitial C-band was observed in some autosome pairs, viz. nos. 2 and 7, among which very conspicuous was a band at the median part of pair no. 7. Two interstitial bands were detected in the X chromosome (Fig. 2B).

4) Silver-stained nucleolus organizers (Ag-NOR): The spiny mouse was

2) G-banding patterns:

All chromosome pairs of the spiny mouse were rather easily identified by the G-band technique (Fig. 4). Figure 3 is a diagrammatic representation of the G-band karyotype. Although the pairs 1 and 2 are apparently similar in lengths, the G-band patterns were quite different between them. Pair no. 1 had 5 conspicuous dark bands, while 3 such dark bands in the pair no. 2. No band was present in the median part of the chromosome no. 2. Pairs no. 3 and 4 were apparently similar in length as well as in banding pattern, while the band in the former was rather complicated than that in

characterized by having a typical Ag-NOR in the following three chromosome pairs; nos. 5, 8 and 12 (Fig. 2C). In pair no. 5 with a short knob the NOR was seen in the short arm of the chromosome. Pair no. 8 showed also Ag-NOR region in the short arm side. In the smallest pair no. 12, the region was observed near the centromere of the long arm, the size of the NOR being sometime heteromorphic.

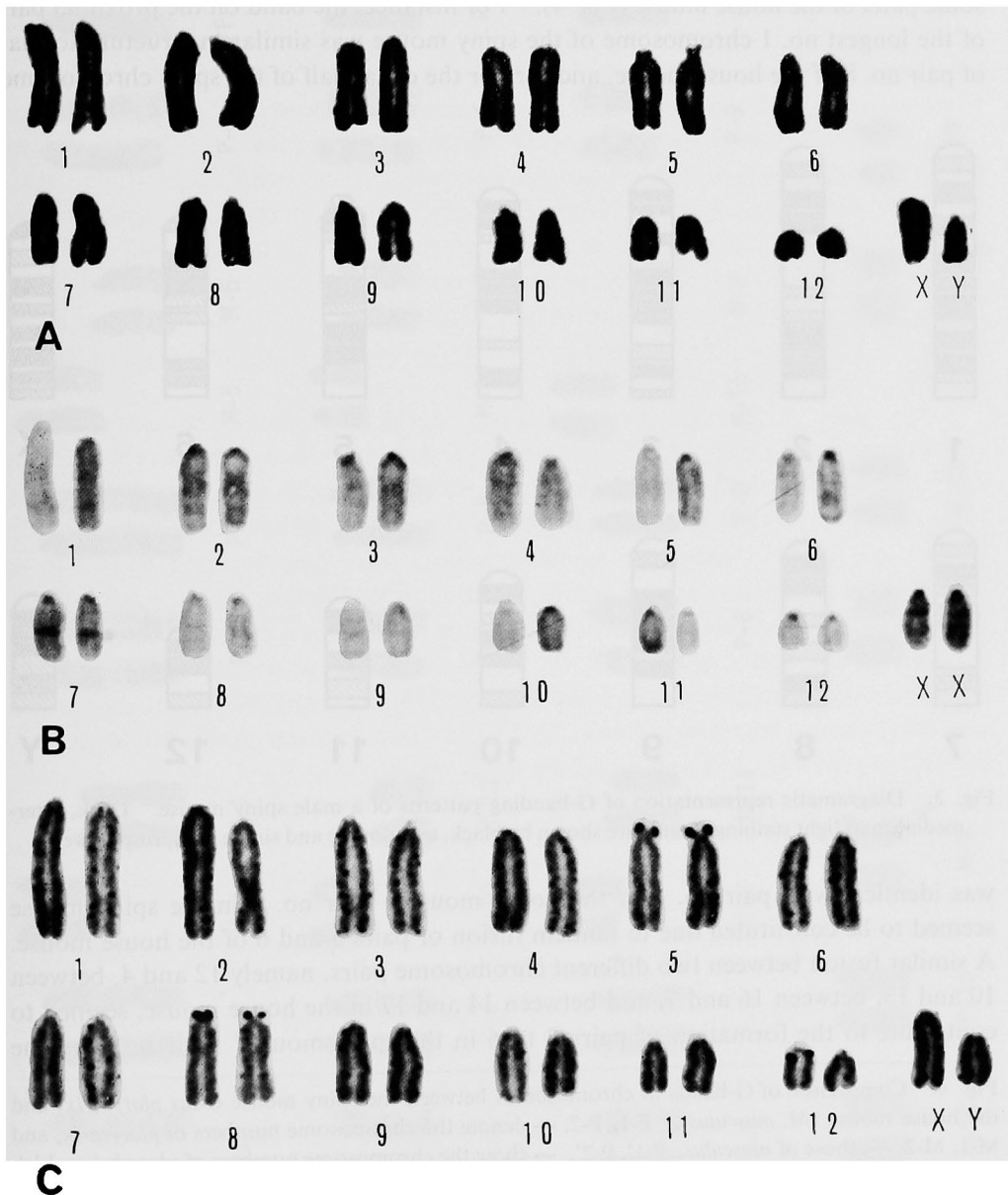


Fig. 2. Karyotypes of the Indian spiny mouse by conventional and differential staining. A, conventional Giemsa staining (♂). B, C-band staining (♀). Bands near centromeres are generally small and unclear, but interstitial bands are seen in some chromosomes. C, Ag-NOR staining (♂). The NOR regions are clearly seen in pair nos. 5, 8 and 12.

II. Comparison of chromosomes between the spiny mouse and the house mouse

1) G-bands: The chromosome numbers were different between the Indian spiny mouse ($2n=26$) and the house mouse ($2n=40$), but their complement consisted of the acrocentric chromosomes. The comparison of the G-bands between the spiny mouse and the house mouse it has revealed that some chromosomes of the spiny were formed as a result of tandem fusion occurring in each set of two chromosome pairs of the house mouse (Fig. 4). For instance, the band on the proximal part of the longest no. 1 chromosome of the spiny mouse was similar in structure to that of pair no. 8 of the house mouse, and further the distal half of the spiny chromosome

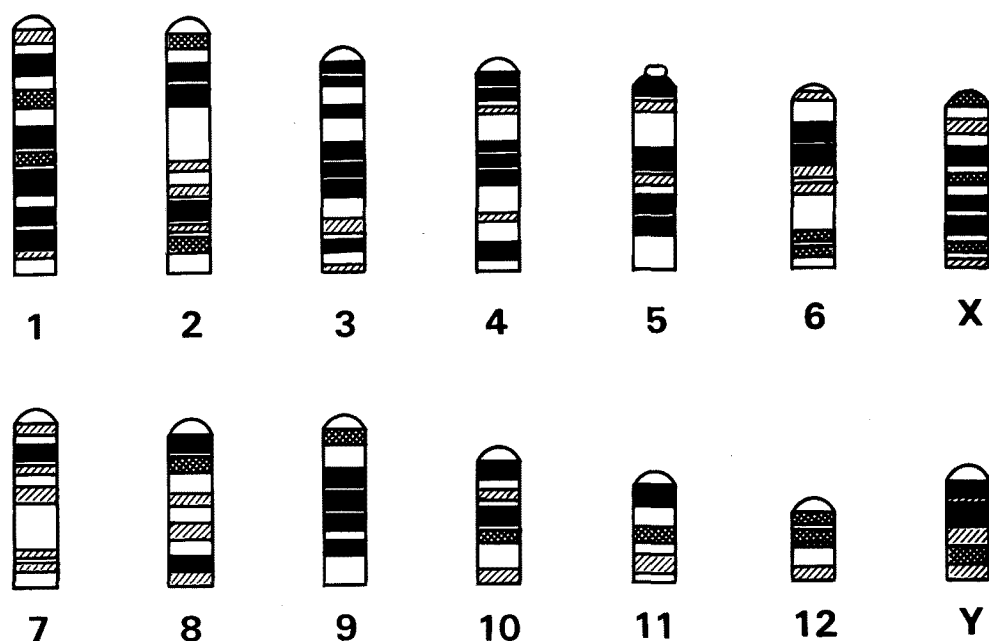
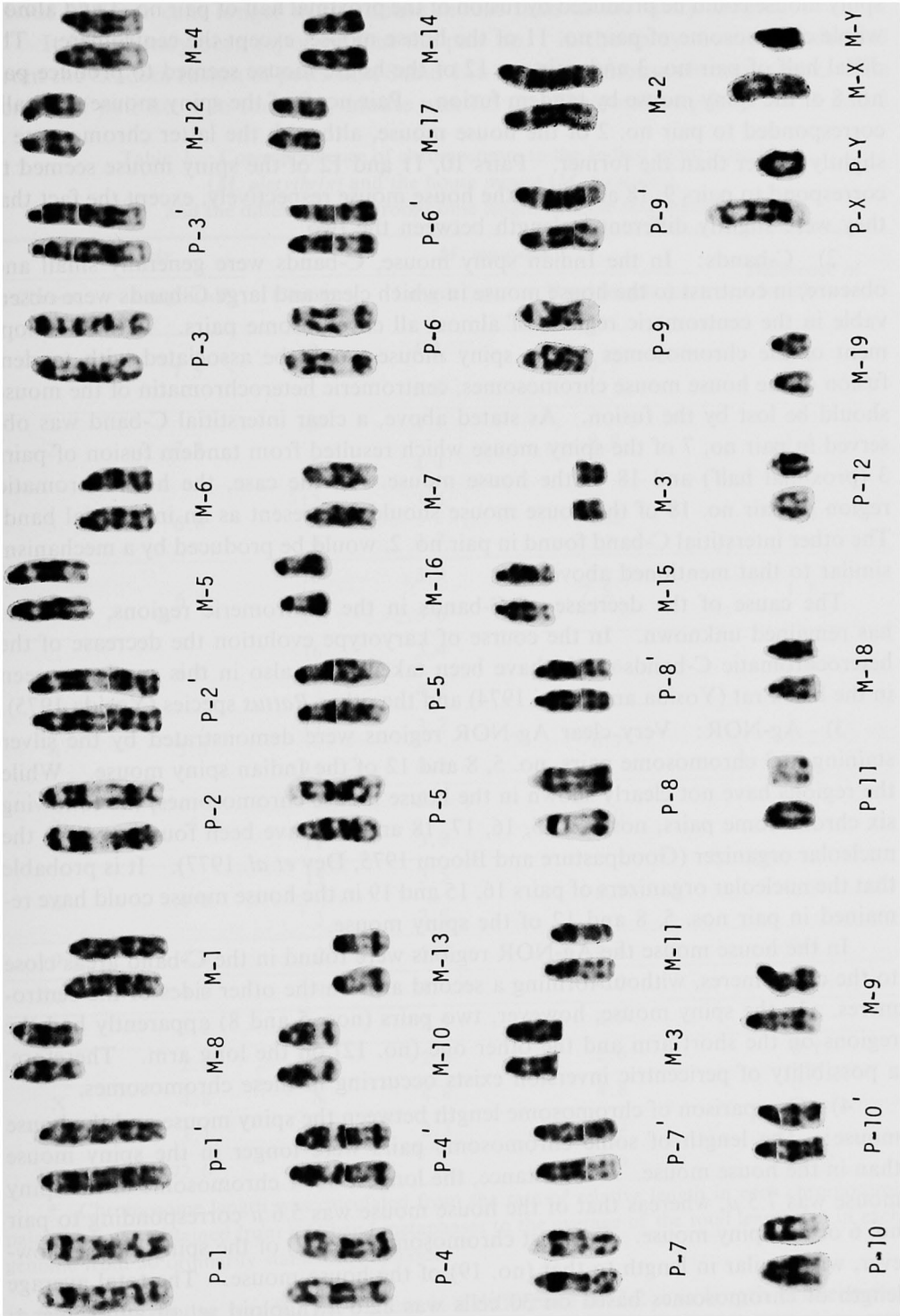


Fig. 3. Diagrammatic representation of G-banding patterns of a male spiny mouse. Dark, intermediate and light stainings bands are shown by black, and double and single lines, respectively.

was identical with pair no. 1 of the house mouse. Pair no. 2 in the spiny mouse seemed to be constituted due to tandem fusion of pairs 5 and 6 of the house mouse. A similar fusion between two different chromosome pairs, namely 12 and 4, between 10 and 13, between 16 and 7, and between 14 and 17 in the house mouse, seemed to contribute to the formation of pairs 3 to 6 in the spiny mouse. Pair no. 7 of the

Fig. 4. Comparison of G-bands in chromosomes between the spiny mouse (*Mus platythrix*) and the house mouse (*M. musculus*). P-1, P-2, --- denote the chromosome numbers of *platythrix*, and M-1, M-2, ---, those of *musculus*. P-1', P-2', --- show the chromosome numbers of *platythrix*, which are artificially composed from *musculus* chromosomes. Pair no. 3 of *musculus* is broken at the middle part, and the two pieces are fused to pair nos. 11 and 15 to make pair nos. 7 and 8 of *platythrix*. Pair nos. 9, 10, 11 and 12 of *platythrix* correspond to pair nos. 2, 9, 18 and 19 of *musculus*, respectively. No. 11 of *platythrix* is derived from the pair no. 9 of *musculus* in which the inversion is taken place. X and Y of *platythrix* are similar to *musculus*.



spiny mouse could be produced by fusion of the proximal half of pair no. 3 and almost whole chromosome of pair no. 11 of the house mouse, except the centromere. The distal half of pair no. 3 and pair no. 12 of the house mouse seemed to produce pair no. 8 of the spiny mouse by tandem fusion. Pair no. 9 of the spiny mouse generally corresponded to pair no. 2 of the house mouse, although the latter chromosome is slightly longer than the former. Pairs 10, 11 and 12 of the spiny mouse seemed to correspond to pairs 9, 18 and 19 of the house mouse respectively, except the fact that they were slightly different in length between the two.

2) C-bands: In the Indian spiny mouse, C-bands were generally small and obscure, in contrast to the house mouse in which clear and large C-bands were observable in the centromeric regions of almost all chromosome pairs. If the development of the chromosomes of the spiny mouse would be associated with tandem fusion of the house mouse chromosomes, centromeric heterochromatin of the mouse should be lost by the fusion. As stated above, a clear interstitial C-band was observed in pair no. 7 of the spiny mouse which resulted from tandem fusion of pairs 3 (proximal half) and 18 of the house mouse. In the case, the heterochromatic region of pair no. 18 of the house mouse should be present as an interstitial band. The other interstitial C-band found in pair no. 2, would be produced by a mechanism similar to that mentioned above.

The cause of the decrease of C-bands in the centromeric regions, however, has remained unknown. In the course of karyotype evolution the decrease of the heterochromatic C-bands might have been taken place also in this species as seen in the black rat (Yosida and Sagai 1974) and the other *Rattus* species (Yosida 1975).

3) Ag-NOR: Very clear Ag-NOR regions were demonstrated by the silver staining in 3 chromosome pairs, no. 5, 8 and 12 of the Indian spiny mouse. While the regions have not clearly shown in the house mouse chromosomes, the following six chromosome pairs, nos. 12, 15, 16, 17, 18 and 19, have been found to have the nucleolar organizer (Goodpasture and Bloom 1975, Dev *et al.* 1977). It is probable that the nucleolar organizers of pairs 16, 15 and 19 in the house mouse could have remained in pair nos. 5, 8 and 12 of the spiny mouse.

In the house mouse the Ag-NOR regions were found in the C-band areas close to the centromeres, without forming a second arm on the other sides of the centromeres. In the spiny mouse, however, two pairs (nos. 5 and 8) apparently had the regions on the short arm and the other one (no. 12) on the long arm. Therefore, a possibility of pericentric inversion exists occurring in these chromosomes.

4) Comparison of chromosome length between the spiny mouse and the house mouse: The length of some chromosome pairs were longer in the spiny mouse than in the house mouse. For instance, the longest no. 1 chromosome of the spiny mouse was 7.5μ , whereas that of the house mouse was 5.6μ corresponding to pair no. 6 of the spiny mouse. Smallest chromosome (no. 12) of the spiny mouse, however, was similar in length to that (no. 19) of the house mouse. The total average length of chromosomes based on 30 cells was 82.6μ (haploid set of autosomes + XY chromosomes) in the house mouse (Table 1). In the spiny mouse the total length of the chromosomes was 32.0μ , about 10.6μ being shorter in the spiny mouse than in the house mouse. Thus it appears that the total length of chromosomes was

about 12.8 per cent longer in the house mouse than in the spiny mouse.

It has already been reported that the pair no. 1 in the spiny mouse was derived from the tandem fusion of pairs nos. 8 and 1 in the house mouse chromosomes. The length of pair no. 1 in the spiny mouse was 7.5 μ . The total length of nos. 1 and 8

Table 1. Correspondence of chromosomes in the Indian spiny mouse (*M. platythrrix*) and the house mouse (*M. musculus*), and the difference of chromosome length in these two species

<i>M. platythrrix</i>		<i>M. musculus</i>			Difference of length (μ)
Chrom. pair	Length(μ)*	Chrom. pair	Length (μ)	Total length (μ)	
1	7.5	8	4.1	9.7	2.2
		1	5.6		
2	7.4	5	4.5	8.9	1.5
		6	4.4		
3	6.8	12	3.5	8.2	1.4
		4	4.7		
4	6.4	10	3.9	7.5	1.1
		13	3.6		
5	6.0	16	3.4	7.6	1.6
		7	4.2		
6	5.5	17	3.2	6.7	1.2
		14	3.5		
7	5.1	3	2.2	6.0	0.9
		11	3.8		
8	4.9	15	3.5	6.2	1.3
		3	2.7		
9	4.8	2	5.2	5.2	0.4
10	4.0	9	4.0	4.0	0
11	3.2	18	2.9	2.9	-0.3
12	2.5	19	2.6	2.6	0.1
X	4.8	X	4.8	4.8	0
Y	3.1	Y	2.3	2.3	-0.8
Total	72.0			82.6	10.6

* Chromosome length was calculated from the rate of relative length in each chromosome pair which was obtained from 8 G-band metaphases to the average of the total length (μ) in each genome from 30 ordinarily stained metaphases.

in the house mouse was 9.7 μ . There is a difference of 2.2 μ in length between the house mouse (pairs no. 1 and 8) and the spiny mouse (pair no. 1). This seems to indicate that 2.2 μ of the chromosomes in the house mouse would have been lost as a result of the tandem fusion leading to the formation of pair no. 1 in the spiny

mouse. In pair no. 2 of the spiny mouse, about 1.5μ in the chromosomes of the house mouse could have been deleted due to the tandem fusion of pair nos. 5 and 6. The chromosome deletion of a similar nature might have occurred in the other chromosome pairs due to the tandem fusion (see Table 1).

It is a general trend that in tandem fusion, some chromosome parts of the house mouse have been lost in relation to the formation of the spiny mouse chromosomes. Thus, the total sum of the deleted chromosome parts in chromosome pairs of the house mouse was 10.6μ . This precisely corresponded to the difference of the total chromosome length between the spiny mouse and the house mouse.

Remarks

Robertsonian fusion-fission mechanism is a common phenomenon for the karyotype evolution in animals in general. Examples of the Robertsonian fusion have been found in the tobacco mouse (*Mus poschiavinus*), the black rats (*Rattus rattus*) and some other rodents. The number of chromosomes in the tobacco mouse was 26 ($n=13$) in which 7 large metacentrics pairs were derived from Robertsonian fusion of 14 acrocentrics (Gropp *et al.* 1970). The black rats are characterized by the chromosome numbers showing $2n=42$, 40 and 38. The variation of the chromosome numbers were assumed to have developed basically due to Robertsonian fusion of 4 acrocentric pairs (Yosida *et al.* 1969, 1974). The Indian spiny mouse (*Mus platythrix*) here dealt with was found to possess the chromosome number quite identical with that of the tobacco mouse, but the chromosome complement consisted of all acrocentrics. Comparative banding analyses of the chromosomes between the spiny mouse and the house mouse have demonstrated that 14 chromosome pairs of the house mouse have been derived from tandem fusion two by two resulting in the formation of the spiny mouse karyotype. Each of pair no. 13 of the house mouse had probably broken in its middle part and then the two pieces thus formed seemed to translocate on pairs nos. 11 and 15, respectively. There is a question whether, or not, the fission of chromosomes had occurred in the spiny mouse chromosomes in relation to the formation of the house mouse karyotype. In the case of fission a new formation of centromeres should be considered. At the present status of investigation, the creation of new centromeres has not been fully demonstrated.

Recent studies have provided a fair example of the tandem chromosome fusion in Indian muntjac, *Muntiacus muntjak*. The latter is characterized by a very low chromosome number as $2n=6$. The situation has been interpreted as a result of the sequential tandem fusion of several chromosome pairs of some other *Muntiacus* species with $2n=46$ (Wurster and Benirschke 1970). In the chromosome of monkeys, Stock and Hsu (1973) described the tandem fusion of centromere and telomeres. Cohen *et al.* (1978) reported a case of the homologous tandem translocation in human chromosomes. Tandem fusion has been recognized as a special case of translocation: it could be defined as the main chromosome part of two or more chromosomes fused tandemly, and thereby one or more centromeres have been lost from the original karyotype to create the new composites. The number of centro-

meres to be lost in the formation of the new karyotype might be shown by a formula of $n/2$ in the case of tandem fusion as occurred in the spiny mouse in which several sets of two chromosomes are involved in the fusion. In the case of sequential tandem fusion as seen in the muntjac, it would be shown by a formula of $n-2$. In these formulae, "n" denotes the number of chromosomes in all pairs involved in the tandem fusion.

Summary

Chromosomes of the Indian spiny mouse (*Mus platythrix*) were observed by conventional, G-, C- and Ag-NOR-staining methods. This species was characterized by 12 acrocentric autosome pairs in addition to acrocentric X and Y chromosomes. The comparison of G-banded chromosomes and chromosome length between the spiny mouse and the house mouse (*M. musculus*) had led the assumption that 8 autosome pairs (nos. 1 to 8) of the former species had been derived from the tandem fusion of two acrocentric pairs of the latter species. Pair nos. 9 to 12 and X and Y in the spiny mouse seemed to correspond in some autosome pairs and X and Y of the house mouse. In this procedure, about 13 percent of the total chromosome length of the house mouse seemed to be lost to produce the new composites.

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